

June 17, 2026

Dear [Admissions Committee],

I am excited to apply for the PhD position "Embodied quantum information flow for collective intelligence in robotic swarms" at the University of Southampton, located in Southampton, United Kingdom. As a passionate AI and Data Science Engineer with a strong background in mathematics and programming, I am confident that my skills align perfectly with the requirements of this project.

My research experience is deeply rooted in the fusion of quantum technologies and swarm intelligence. In my previous work on "Cardiovascular Medical Digital Twin", I developed a linear-time state space model for sub-millisecond bedside inference (Medical_{MambaTS}). *This achievement showcases my ability to create efficient algorithms that can run in real-time, which is crucial for the development of collective robotics systems.*

Furthermore, my experience with physics-informed neural networks (PI_{DeepONet}) using PyTorch Autograd has allowed me to

I am particularly drawn to this project because it offers a unique opportunity to bridge the gap between quantum technologies and embodied swarm intelligence. My background in AI and Data Science, combined with my research experience in robotics and machine learning, makes me an ideal candidate for this position.

I look forward to contributing my skills and expertise to the University of Southampton's research community.

Sincerely, Ismail Aissaoui